Ch. 22 HW #4 Solutions

Principles of Microeconomics

November 14, 2025

QuickQuiz

8. (c)

   Lobster:

   - Substitution effect: Negative because lobster becomes relatively more expensive.
   - Income effect: Negative because lobster is a normal good so consumption decreases when purchasing power decreases.
   - Overall effect: Negative because the substitution and income effects are both negative.

   Chicken:

   - Substitution effect: Positive because chicken becomes relatively cheaper.
   - Income effect: Positive because chicken is inferior so consumption increases when purchasing power decreases.
   - Overall effect: Positive because the substitution and income effects are both positive.

9. (c)

   - Substitution effect: Negative because pasta becomes relatively more expensive.
   - Income effect: Positive because the overall effect is positive, so the income effect must be positive to counteract the substitution effect.
   - If the income effect is positive in response to a decrease in purchasing power, pasta must be an inferior good.

10. (d) – For the labor supply curve to slope upwards, labor must increase in response to an increase in wage. Since we assume all of a person's time is spent on labor or leisure, an increase in labor is the same thing as a decrease in leisure. We can think about the income and substitution effect of an increase in wage on demand for leisure to answer this question.

- Substitution effect: If wage increases, leisure becomes relatively more expensive. As a result, the substitution effect is negative. That is, Maude decreases her demand for leisure.
- Income effect: If wage increases, Maude effectively becomes richer. Leisure is a normal good, so the income effect is positive. That is, Maude increases her demand for leisure.
- Overall effect: For labor supply to increase, her demand for leisure must decrease on the whole. That means the substitution effect must be greater than the income effect.

11. (b)

- Substitution effect: When the interest rate falls, you have to save more money when you're young in order to consume a given amount when you're old. To save more, you have to consume less when young. Therefore, the opportunity cost of consumption when old increases. That's to say, consumption when old becomes relatively more expensive when the interest rate falls. As a result, the substitution effect is negative.
- Income effect: When the interest rate falls, your savings generate less money. That makes you effectively poorer. Consumption when old is a normal good, so the income effect is negative.
- Overall effect: Negative because the substitution effect and income effect are both negative.

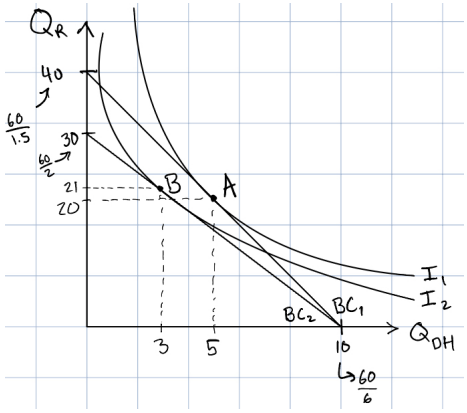
Problems & Applications

6. (a) True – When the price of a Giffen good increases, quantity demanded of the Giffen good increases. A price increase makes the Giffen good relatively more expensive, so the substitution effect is negative. The overall effect of a price increase is positive, so the income effect must be positive to counteract the negative substitution effect. A positive income effect implies that a decrease in purchasing power causes an increase in consumption. That means Giffen goods must be inferior.

   (b) False – Almost all inferior goods are not Giffen goods. When the price of a typical inferior good increases, the substitution effect outweighs the income effect. The decrease in consumption from the change in relative prices outweighs the increase in consumption from the change in purchasing power. We know that because few if any goods have upward sloping demand curves.

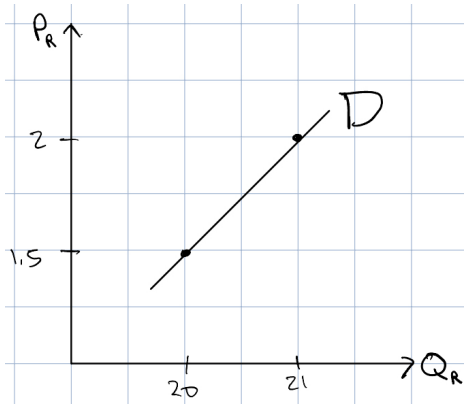
7. (a) Priya's budget constraint is $60 = 6Q_{DH} + 1.5Q_R$. Since she spends an equal amount on both goods, she spends \$30 on each. That means she buys 5 dining-hall meals and 20 cups of Ramen. See plot below.

(b) Now her budget constraint is $60 = 6Q_{DH} + 2Q_R$. 30% of her budget is \$18, so she buys 3 dining-hall meals. She must spend the remaining \$42 on Ramen, so she buys 21 cups of Ramen.

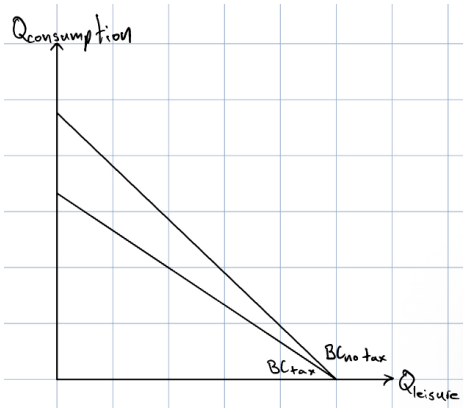


(c) Her Ramen consumption increased by 1. We know the substitution effect is negative because the price increase made Ramen relatively more expensive. Since the overall effect was positive, her income effect must have been positive and larger than the negative substitution effect.

(d) It's a Giffen good because quantity demanded increases when price increases.

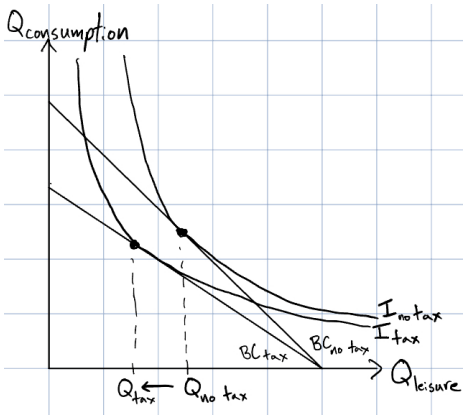


8. (a)

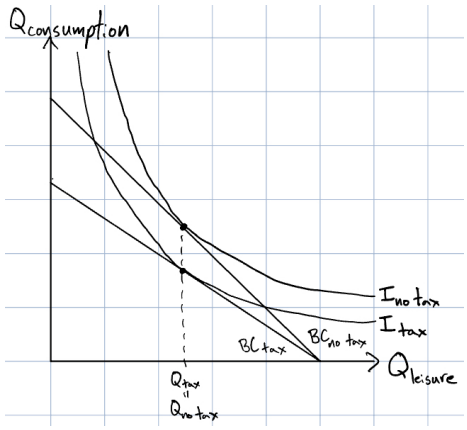


(b) The tax makes consumption more expensive because you have to work more hours to consume at a certain level. That means leisure is relatively less expensive, so the substitution effect on leisure is positive. The tax decreases your income, and leisure is a normal good, so the income effect is negative. Whether or not you work more hours, fewer hours, or the same number of hours depends on the relative size of the substitution effect and income effect.

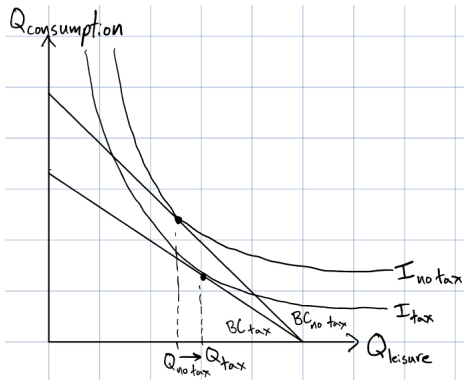
If the income effect is larger than the substitution effect, you will engage in less leisure and work more hours:



If the income effect is equal to the substitution effect, you will engage in the same amount of leisure and work the same number of hours:



If the income effect is smaller than the substitution effect, you will engage in more leisure and work fewer hours:



9.

